

Can AI help secure 2.5 million small businesses?

Taking Australian Small Business from Essential Eight Awareness to Action with AI-Assisted Support · Rick Taylor (11870153B)

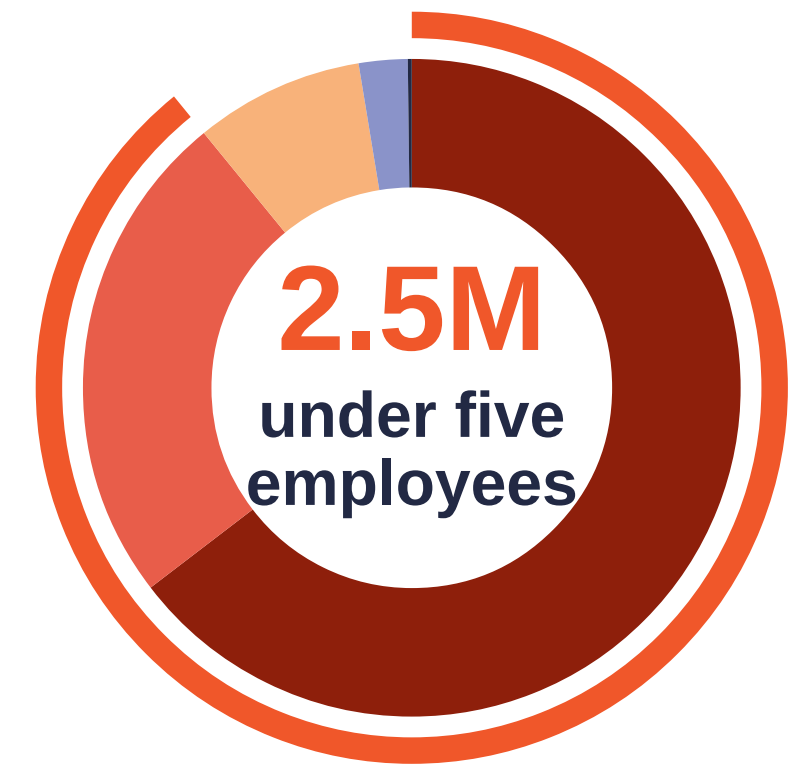
ITC571 Emerging Technologies and Innovation · Stream B: Cybersecurity, Networking, Privacy, Risk, Ethics

Abstract Most small businesses have no one with IT skills while AI speeds up attacks. From 28 sources, this proposal frames design principles for AI help owners can rely on.

INTRODUCTION & BACKGROUND

1 "Small business" is mostly one person

All 2,814,778 Australian businesses, by number of employees, June 2026



ABS, 2026 (Counts of Australian businesses, Table 13)

Risk area: under five · 89%

64.6% no employees 1,818,575
24.5% 1-4 employees 689,600

Everyone else · 11%

8.3% 5-19 232,912
2.4% 20-199 68,325
0.2% 200+ 5,366

2.5M owners,
no IT team

INTRODUCTION & BACKGROUND

2 Faster attacks, bigger losses

+11%
incidents ASD
responded to

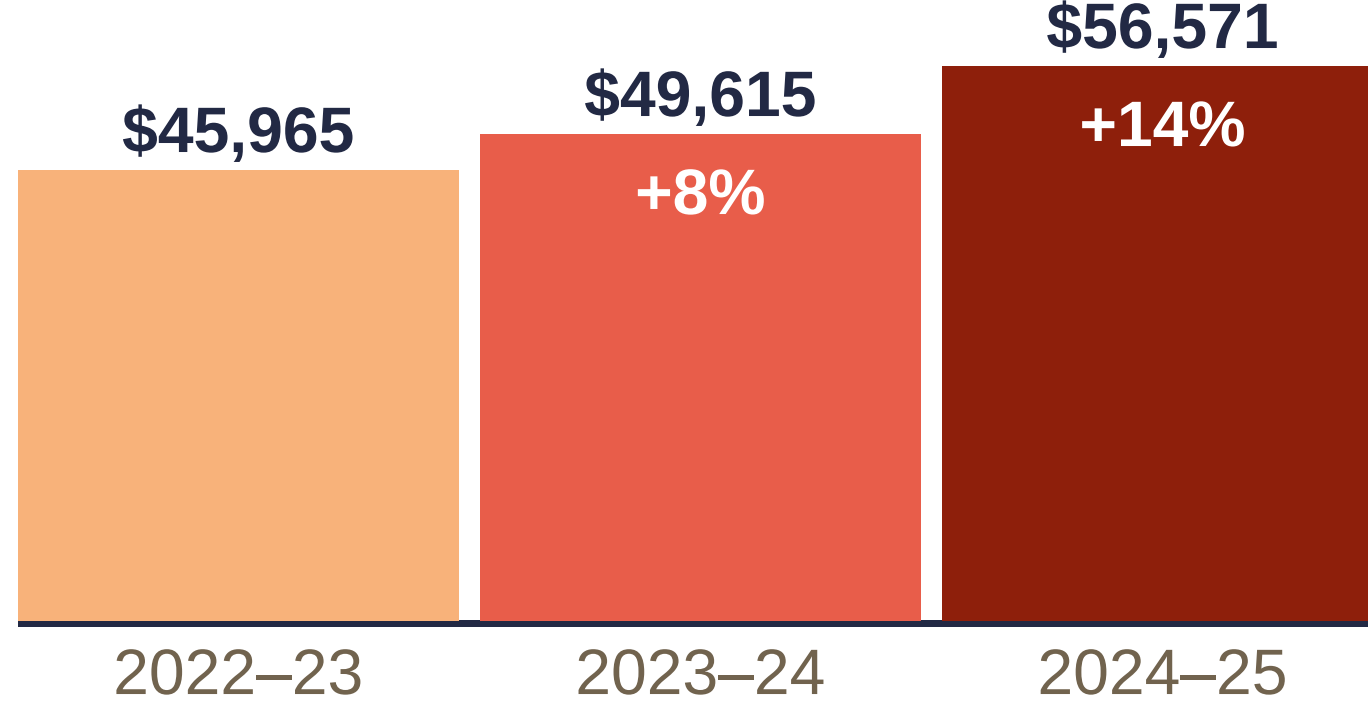
2024-25 vs the previous year (ASD, 2025)

+83%
ASD warnings of
attacks

AI "almost certainly" enables attacks "on a larger scale and at a faster rate"

ASD, 2025, p. 10

Average cost per cybercrime report, small business (ASD, 2025)



EXPECTED · RESEARCHER'S VIEW

AI lowers the entry point for attackers and widens the net

KEY FINDINGS & DISCUSSION

3 Help exists. Why isn't it working?

PRELIMINARY

Essential Eight ACSC, 2023

× A specification, not an explanation
× No certificate, no point of completion

40+ technical rules at that level (derived from ACSC, 2023)

99.8%

of businesses are pointed at one target, Maturity Level One

ASD, 2024; ABS, 2026

SMB1001 DSI, 2025

× Must be bought before it can be read

Cyber Wardens COSBOA, 2026

× Awareness, not implementation

Paid support Khan et al., 2025; Wilson et al., 2023

× Rarely sought; owners self-research

0 of 28 sources measure trust or time as a barrier for owners: the gap this study opens

Coding pass, 28 sources

INTERPRETATION

Awareness without action: **3 in 4** owners find the information too technical

WA survey, n = 36 · Chidukwani et al., 2024

CONCLUSION & FUTURE WORK

5 Conclusion: contribution & future work

Contribution

- ✓Mapped the gap: only 2 Australian peer-reviewed studies; trust and time unstudied
- ✓Proposed chain: capability first, cost last
- ✓A testable hypothesis: assist, don't act

Future work: the data that's missing

- ? Trust and time as barriers
- ? Do owners see themselves as targets? Does AI change that?
- ? How secure are they, beyond self-report?
- ? What do owners rely on to judge advice?

Ask me: why is cost the last barrier, not the first?

4 Can AI close the gap?

Aim Design principles that match an owner's reliance to what the AI can actually do, without disrupting the business.

Objectives explain why E8 stalls · sort which barriers design can reach · name the risks AI adds · specify how to measure reliance Raees & Papangelis, 2026

OVERARCHING RESEARCH QUESTION

What design principles make an AI agent trustworthy to someone with no basis for judging it?

AI as the lifeline?

- RQ1** Why has the Essential Eight, despite government backing, not improved small business security practice?
- RQ2** Which of the barriers beyond the framework can the design of support address?
- RQ3** How does an owner form trust in security advice where they have no knowledge of the subject?
- RQ4** Does support that assists the owner address the barriers better than support that acts for them?
- RQ5** What new risks does an AI agent introduce that were not previously present?

WORKING HYPOTHESIS · the owner can't check AI's work, so AI must assist, not act
AI could identify · explain · sequence · check its own work

→ only the owner can **CONSENT**

RESEARCH METHODOLOGY & METHODS

Method: Literature Analysis Model · no participants

4 search strings
Primo · Scholar · AI-assisted

28 sources
15 peer-reviewed · 13 other

7 axial codes
EndNote, open then axial

Themes · next
reflexive thematic analysis



REFERENCES
ricktaylor1-cell.github.io/
itc571-a3-poster/
references